

Energy yield of bifacial textured perovskite/silicon tandem photovoltaic modules

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ABSTRACT

Bifacial perovskite/crystalline silicon (c-Si) tandem photovoltaic (PV) modules that harvest albedo radiation are a promising strategy to further enhance the energy yield (EY) of monofacial perovskite/c-Si tandem PV modules. Given the required current matching in bifacial perovskite/c-Si two-terminal (2T) tandem PV modules, predicting the expected enhancement in EY requires an advanced EY modelling framework. In this study, the architecture, module tilt, perovskite absorber layer thickness, and bandgap of the perovskite top solar cell are optimized for several types of grounds and two locations. Compared to bifacial c-Si single-junction PV modules, the relative enhancement in EY of textured bifacial perovskite/c-Si 2T tandem PV modules is around 24–38% for an exemplary grass ground (mean albedo = 35%). With increasing albedo, the optimum bandgap of the perovskite top solar cell decreases from 1.72 eV (black ground, mean albedo = 0%) to 1.55 eV (snow, mean albedo = 88%). This is attributed to enhanced current generation only in the c-Si bottom solar cell due to albedo radiation, entering the solar cell at the rear side. In addition, minor optical losses due to ground shading are found for modules with a finite mounting height in the range of 1 m. Overall, our study highlights the importance of EY modelling to assess the performance of bifacial perovskite/c-Si tandem PV. It provides direction for the design of bifacial perovskite/c-Si tandem solar cells with regard to the device architecture and the choice of the perovskite material with regard to its bandgap.

1. Introduction

After only four years of development, perovskite/silicon tandem photovoltaics (PV) has already demonstrated high power conversion efficiencies (PCEs), exceeding the PCE of crystalline silicon (c-Si) single-junction (SJ) PV [1]. Given the tunable bandgap and inexpensive fabrication techniques [2–4], perovskite PV enables a cost-effective way to advance the PCE of state-of-the-art c-Si SJ PV modules [5]. Bifacial PV modules promise a route to further enhance the energy yield (EY) by harvesting the photons incident on the rear side of the PV module [6–8]. Bifacial PV modules typically employ cover glasses at both, front and rear side, and a transparent rear electrode, enabling the diffuse and direct light reflected/scattered at the ground, namely albedo, to contribute to the power generation [9,10]. The concept is widely applied to SJ solar cells [11–13], but conceptually also very promising for

tandem solar cells [14].

To date, two architectures of different electrical configurations of perovskite/c-Si tandem solar cells are investigated – the two-terminal (2T) configuration as well as the four-terminal (4T) configuration. The 2T configuration uses a monolithic interconnection of a semitransparent perovskite top solar cell and a c-Si bottom solar cell with only two electrodes, resulting in low optical losses [15]. This approach, however, also requires a layer stack optimization in order to minimize the power loss due to current mismatching in the sub-cells [1,16]. In contrast, the 4T configuration consists of a mechanically-stacked perovskite solar cell on top of a c-Si solar cell, enabling independent operation of the sub-cells. In this configuration, three or four electrodes are required, inducing enhanced parasitic optical absorption in the contact layers [17–19]. For monolithic perovskite/c-Si 2T tandem solar cells a record PCE of 28% has been certified [1].

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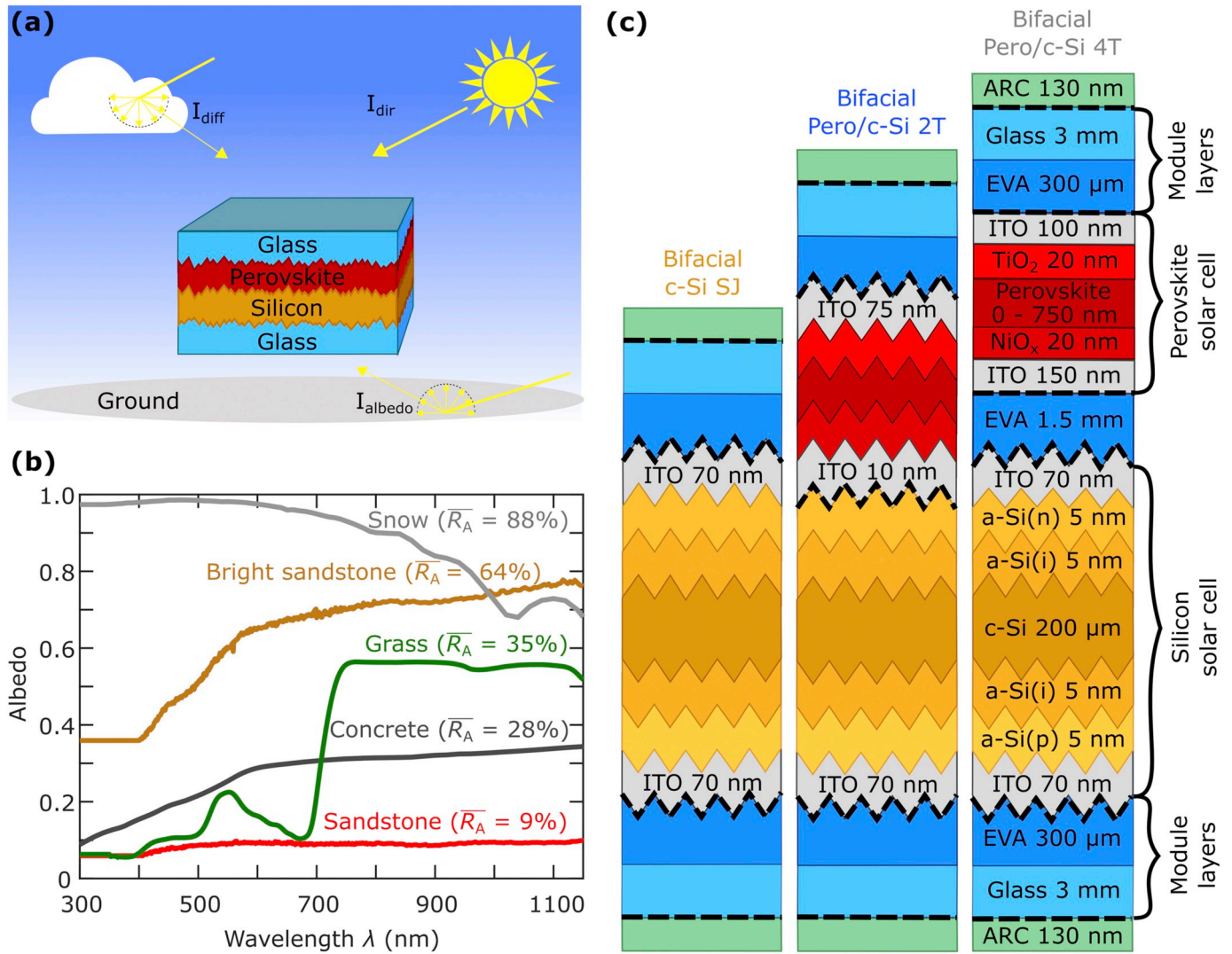


Fig. 1. (a) Sketch of the direct, diffuse, and albedo irradiance on a bifacial PV module. (b) Reflection spectra of grounds used for the simulation of albedo irradiance. The data is taken from the ecosystem spaceborne thermal radiometer experiment on space station (ECOSTRESS) spectral library [38]. Additionally, the mean albedo ($\bar{R}_A$) of each ground in the wavelength range of 300–1150 nm is specified in the legend. (c) Three architectures of bifacial PV modules used for the calculation of EYs. (1) Bifacial c-Si SJ PV module with micron-scale pyramidal c-Si texture at the front and rear. (2) Bifacial perovskite/c-Si 2T tandem PV module with c-Si texture at the front and rear. (3) Bifacial perovskite/c-Si 4T tandem PV module with planar semitransparent perovskite top solar cell and c-Si texture at the front and rear of c-Si bottom solar cell. The layer stack of the c-Si bottom solar cell is indium tin oxide (ITO), p-doped amorphous Si (a-Si(p)), intrinsic a-Si (a-Si(i)), c-Si, a-Si(i), and n-doped amorphous Si (a-Si(n)). The perovskite top solar cell consists of ITO, nickel oxide (NiO_x), perovskite, titanium dioxide (TiO_2), and ITO. In order to reduce reflection losses, an anti-reflective coating (ARC) is included on top of the glass at the front and rear side of the PV modules. In case of the bifacial perovskite/c-Si 4T tandem PV module, the encapsulant ethylene-vinyl acetate (EVA) is used in between the top and bottom solar cell.

Light management is of key importance in perovskite/c-Si tandem solar cells in order to maximize the light in-coupling, e.g., by reducing front-side reflection losses [15]. Significant PCE improvements were achieved with textured foils for light management at the front side of the tandem solar cell [16]. The use of random micron-scale textures at the front and rear side of c-Si strongly enhances the PCE of perovskite/c-Si tandem solar cells by light in-coupling and trapping [20]. Random pyramidal c-Si textures are commonly fabricated by an inexpensive etch step of the c-Si wafer [21]. Remarkable work on semitransparent perovskite SJ solar cells for the application in perovskite/c-Si 4T tandem solar cells, led to PCEs of up to 27.1% [17,19].

Determining the EY of a tandem solar cell has to account for realistic outdoor conditions. Given the strong variations of the incident sun spectrum and incident angle over time, it is of decisive importance to optimize the layer stack of perovskite/c-Si 2T tandem PV modules with regard to EY, in order to reduce the inevitable current matching losses in the sub-cells. EY modelling was successfully applied previously to assess

monofacial and bifacial c-Si SJ PV modules [22–25]. Moreover, it was used to determine optimal bandgaps, comparisons of architectures and installation angles of monofacial perovskite/c-Si tandem PV modules and perovskite/CIGS tandem solar cells in 4T as well as 2T configuration [26–31]. So far, there are numerical calculations for bifacial perovskite/c-Si tandem PV modules as well as c-Si SJ PV modules, investigating the device performance under normal light incidence with constant front- and rear-side illumination [10,32,33]. But there is no previous report on EY modelling of bifacial perovskite/c-Si tandem PV modules. For this architecture, EY modelling is particularly important, since the albedo irradiance is mainly harvested in the c-Si bottom solar cells, which is expected to significantly impact the optimization of bifacial perovskite/c-Si 2T and 4T tandem PV modules.

In this work, we study the EY of bifacial textured perovskite/c-Si 2T and 4T tandem PV modules for various albedos arising from different grounds: sandstone, concrete, grass, bright sandstone, and snow. For comparison, also the EY of monofacial perovskite/c-Si 2T tandem PV

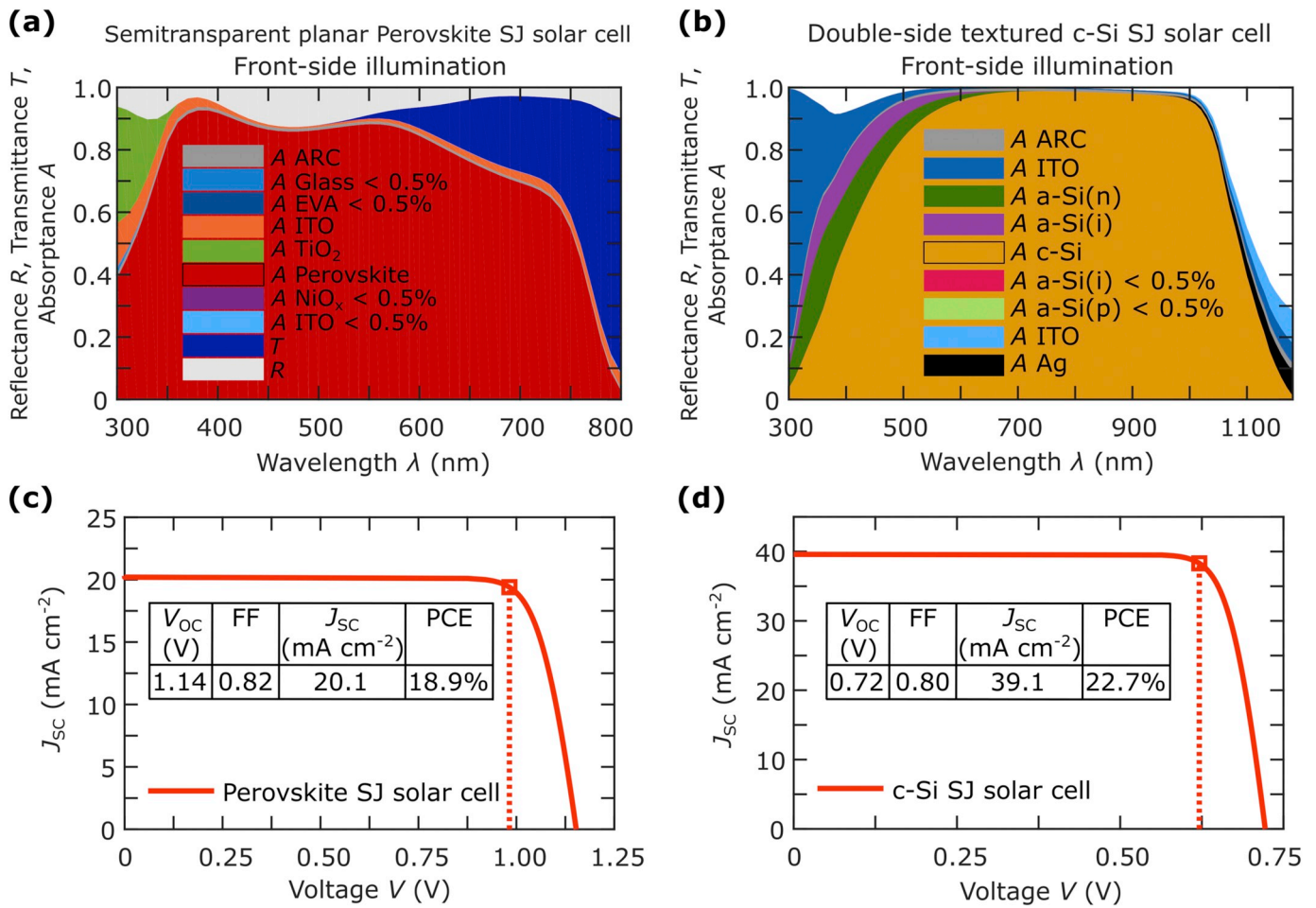


Fig. 2. Reference devices: (a) Optical analysis of the semitransparent planar perovskite SJ solar cell, used for the simulation of perovskite/c-Si tandem PV modules. The device architecture is ITO (150 nm)/ NiO_x (20 nm)/Perovskite (400 nm)/ TiO_2 (20 nm)/ITO (100 nm)/EVA (300 μm)/Glass (3 mm)/ARC (90 nm). Here, the perovskite absorber is methylammonium lead iodide (MAPbI_3) with a bandgap of 1.55 eV. (b) Corresponding analysis of the opaque double-side textured c-Si SJ solar cell with the device architecture Ag (100 nm)/ITO (70 nm)/a-Si(p) (5 nm)/a-Si(i) (5 nm)/c-Si (200 μm)/a-Si(i) (5 nm)/a-Si(n) (5 nm)/ITO (70 nm)/ARC (60 nm). For the semitransparent perovskite SJ solar cell, the overall transmittance (T) and overall reflectance (R) is calculated in addition to the absorbance (A) of each layer. (c–d) Electrical properties of both solar cells. The presented values of PCE, V_{OC} , fill factor (FF), and J_{SC} in the inset figure are rounded for reasons of clarity.

modules as well as monofacial and bifacial c-Si SJ PV modules is determined. Our study shows that in contrast to bifacial perovskite/c-Si 4T tandem PV modules, the optimal bandgap of perovskite in bifacial perovskite/c-Si 2T tandem PV modules is strongly influenced by albedo. As we highlight, this is due to current matching in the sub-cells, since the gain of short-circuit current density (J_{SC}) in the c-Si bottom solar cell due to albedo radiation can be compensated by adjusting the bandgap of the perovskite absorber layer. We find that with increasing albedo significantly lower bandgaps are needed for optimal device performance. We quantify the very remarkable gain of bifacial perovskite/c-Si tandem PV modules to be 27–43% for common grounds like concrete (mean albedo = 28%), compared to bifacial c-Si SJ PV modules. Finally, the influence of ground shading on the EY of bifacial perovskite/c-Si 4T tandem PV modules is studied in order to quantify losses in the power output for a realistic mounting height of the PV modules above the ground.

2. Methods

In this study, the performance of bifacial perovskite/c-Si tandem PV modules under realistic irradiation conditions is investigated by means of numerical methods. For the evaluation of EY, we use an in-house developed software for arbitrary tilted bifacial PV modules and for locations with different climate zones, e.g., hot desert climate and

temperate climate [34]. Details on the numerical method have been recently published by Schmager et al. [35]. The simulation platform allows fast and precise EY calculations of perovskite and Si based multi-junction solar cells. For perovskite/c-Si tandem multi-junction solar cells in 2T- and 4T-configuration, the annual EY is determined by using a temperature-dependent one-diode model mimicking the electrics of each sub-cell.

The incident angular- and spectral-dependent direct and diffuse sky irradiance is calculated from hourly-resolved typical meteorological year (TMY3) data by combining a simple cloud model and the established model of atmospheric radiative transfer of sunshine (SMARTS) [36,37]. By using spectral reflection data of the ecosystem spaceborne thermal radiometer experiment on space station (ECOSTRESS) spectral library [38,39], the spectral albedo irradiance is determined from the sky irradiance. In Fig. 1a, the irradiance on a bifacial PV module is schematically illustrated. The reflection spectra of all grounds used in this study are shown in Fig. 1b. The spectral albedo intensity of all grounds is indicated with the arithmetic mean albedo reflectance $\bar{R}_A$ for better legibility. We assume Lambertian scattering of the incident sky irradiance at the ground and calculate the spectral albedo irradiance from the reflection spectra with input of the incident direct and diffuse irradiance.

In order to assess the EY for elevated bifacial PV modules, which

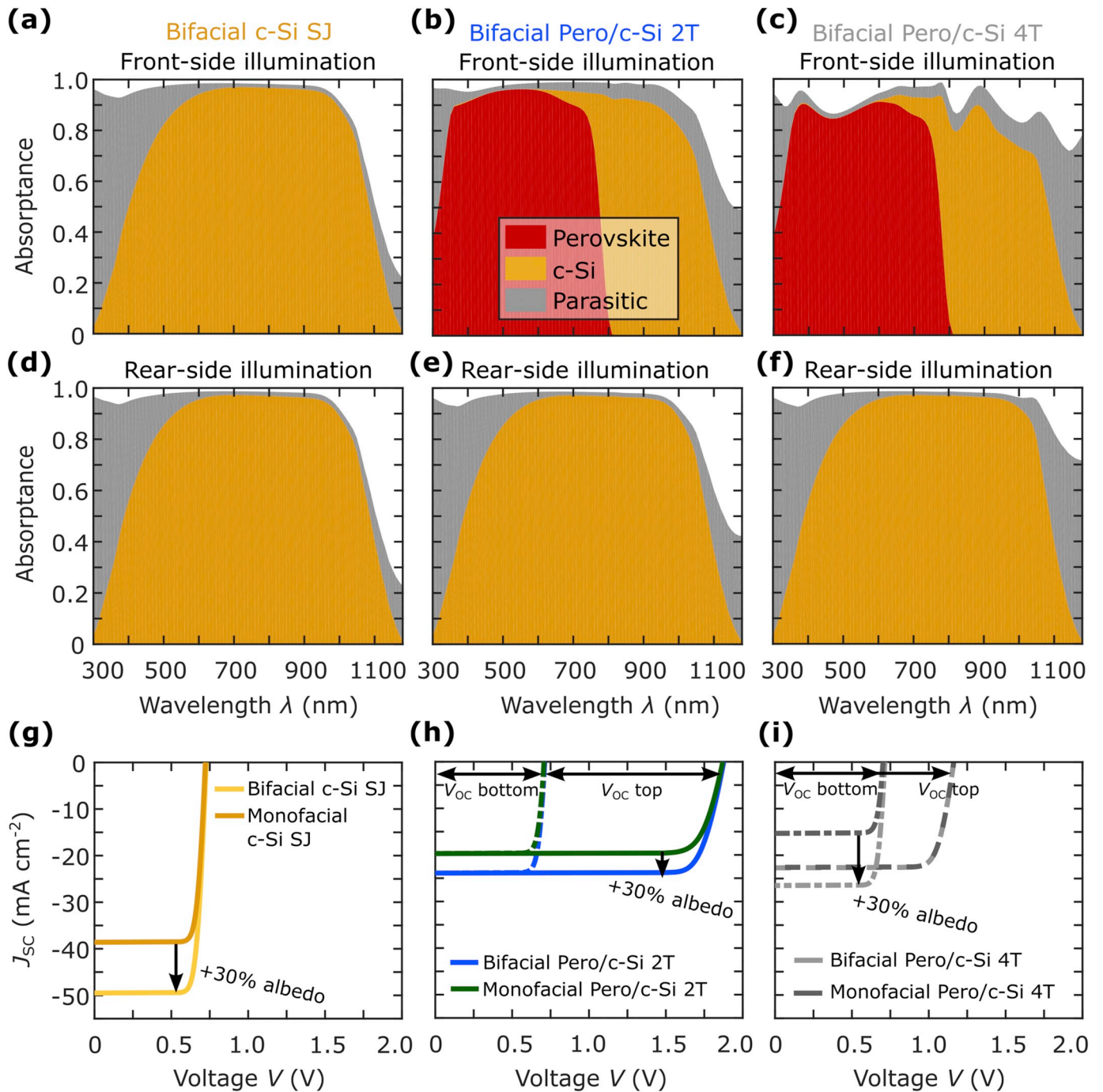


Fig. 3. Optical analysis of the architectures of bifacial PV modules as illustrated in Fig. 1b, for (a–c) front-side illumination as well as for (d–f) rear-side illumination. Here, the perovskite absorber exhibits a bandgap of 1.55 eV. The absorbance is split into perovskite absorber layer, crystalline c-Si absorber layer, and overall parasitic absorption in the other layers. (g–i) Corresponding calculated $J-V$ characteristics for illumination with air-mass 1.5 global (AM1.5 g) spectrum and normal incidence at front side and additional illumination with 30% intensity of AM1.5 g with normal incidence at the rear side. In order to illustrate the gain in power output due to rear-side illumination, the $J-V$ characteristics of all corresponding monofacial architectures is also shown.

mainly suffer from ground shading, we implemented a method for determining the resulting loss in EY according to the method of Sun et al. [7]. The remaining fraction of incident light from ground is commonly calculated by deriving view factors from geometrical view angles describing illuminated and shadowed areas on the ground [40–42]. The model considers the ground shading for direct and diffuse incident light, since the shadow cast differs with the type of irradiance. Hence, the reflected irradiance is separated into reflected direct and reflected diffuse irradiance in order to calculate the remaining albedo irradiance.

For the calculation of reflected direct irradiance, the view factor for the shaded area below the PV module is deduced from the shadow width at ground, according to Hottel et al. [40]. For the calculation of reflected diffuse irradiance, a directional view factor is determined as function of mounting height and width of the PV module.

By means of optical simulation, the angular and spectral reflectance, transmittance, and absorbance in the layer stack of the semitransparent devices is determined for front-side as well as rear-side illumination. The optical model combines a transfer-matrix method for thin layers and ray

optics for optically thick layers in order to calculate the light propagation throughout the complex layer stack of bifacial perovskite/c-Si tandem PV modules [35]. The reflection and transmission of light at the textured c-Si surfaces is treated by geometrical ray tracing according to the method of Baker-Finch & McIntosh [43].

The investigated architectures of perovskite/c-Si tandem PV modules are shown in Fig. 1c. These architectures feature micron-scale textures at both the front and rear sides of the c-Si wafer, referred to as a double-side texture throughout the remainder of this work. In case of 4T configuration, the architecture consists of a planar perovskite SJ solar cell on top of a double-side textured c-Si SJ. The reference devices are monofacial and bifacial state-of-the-art c-Si SJ PV modules as well as monofacial perovskite/c-Si 2T tandem PV modules. In order to prove that these devices serve as references, in Fig. 2, we show the optical and electrical properties of the simulated perovskite SJ solar cell as well as c-Si device – a heterojunction with intrinsic thin layers (HIT) – under standard test conditions (STC) [44]. These solar cells exhibit a typical architecture and PCE of commonly used solar cells for the application in perovskite/c-Si tandem PV [45,46]. The investigated architectures of perovskite/c-Si tandem PV modules are based on these reference devices. For bifacial perovskite/c-Si 2T tandem PV modules, the EY of further conceivable architectures was also determined. These architectures are: (i) c-Si with planar front and rear side, (ii) c-Si with planar

front and textured rear side, (iii) front glass texture and c-Si with planar front and textured rear side, and (iv) the above-mentioned c-Si with double-side texture (see Fig. S1, Supporting Information). The reference monofacial perovskite/c-Si 2T tandem PV module features double-side c-Si texture. The impact of employing light management textures in monofacial perovskite/c-Si 2T tandem PV modules was studied before, quantifying the EY also for different architectures [30]. In order to deduce reasonable cost-value ratios for marketable PV, we further calculate the EY of bifacial and monofacial c-Si SJ devices with double-side texture as reference and identify the gain in EY of bifacial perovskite/c-Si tandem PV modules.

3. Results

First of all, an optical analysis of the introduced architectures of bifacial PV modules is performed with normal light incidence in order to investigate the effect of albedo irradiance on the absorption and hence current generation in bifacial devices. Fig. 3a-f shows the calculated absorptance of the reference bifacial c-Si SJ PV module for front- and rear-side illumination, as well as the absorptance of both bifacial perovskite/c-Si 2T and 4T tandem PV modules. Here, the bandgap of the perovskite absorber layer is 1.55 eV. With respect to current matching in the bifacial perovskite/c-Si 2T tandem PV module, the layer stack is optimized for the exemplary case of albedo irradiance with a ground reflection of 30%. The architecture of bifacial perovskite/c-Si 4T tandem PV module shows distinct reflection losses for front-side illumination due to the planar top solar cell, and low reflection losses for rear-side illumination due to improved light in-coupling due to the c-Si texture. In the case of rear-side illumination, the in-coupled light is fully absorbed in the low bandgap c-Si absorber of bifacial perovskite/c-Si tandem PV modules. The absorptance in the corresponding monofacial perovskite/c-Si 2T and 4T tandem PV modules is shown in Fig. S2 (Supporting Information). For front side illumination, the overall absorptance in the monofacial tandem PV modules is slightly increased, since in bifacial PV modules a fraction of the infrared light is transmitted. For rear side illumination no light enters the solar cell due to the opaque metal electrode and hence no light is absorbed.

In order to demonstrate the effect of rear-side illumination on the device performance of bifacial perovskite/c-Si tandem PV modules, we additionally determine the current-density versus voltage (J - V) characteristics for the investigated architectures. Accordingly, we show in Fig. 3g-i the current generation in bifacial as well as monofacial PV modules. The calculation of J_{SC} for front-side illumination is performed according to STC. Summing up the absorptance for front- and rear-side illumination in bifacial PV modules, the resulting J_{SC} in the c-Si bottom solar cell of all architectures compared to the monofacial PV modules is essentially increased by the albedo radiation, while J_{SC} in the perovskite top solar cell is not enhanced. The radiation from the rear side is almost completely absorbed in the c-Si bottom solar cell. Hence, in case of bifacial perovskite/c-Si 2T tandem PV modules the perovskite absorber layer thickness needs to be adapted in order to optimize the total J_{SC} for any given albedo intensity. The corresponding J - V characteristics of the perovskite/c-Si 2T and 4T tandem PV modules for different light intensities with air-mass 1.5 global (AM1.5 g) spectrum are shown in Fig. S3 (Supporting Information).

Having investigated the performance of bifacial perovskite/c-Si tandem PV modules under STC, in a subsequent analysis these devices are evaluated with regard to their EY. Under realistic irradiation conditions, the angle of incidence, intensity, and spectrum of sky irradiance incident on a PV module change during the course of a day as well as the seasons, and therefore the effects on the device performance as well as on the optimal design has to be studied. One key question is the influence of albedo irradiance on the EY of bifacial perovskite/c-Si 2T tandem PV modules. In Fig. 4, we show an increase of the optimal perovskite absorber layer thickness with increasing albedo irradiance in order to obtain maximal EY. Firstly, the EY of the reference monofacial

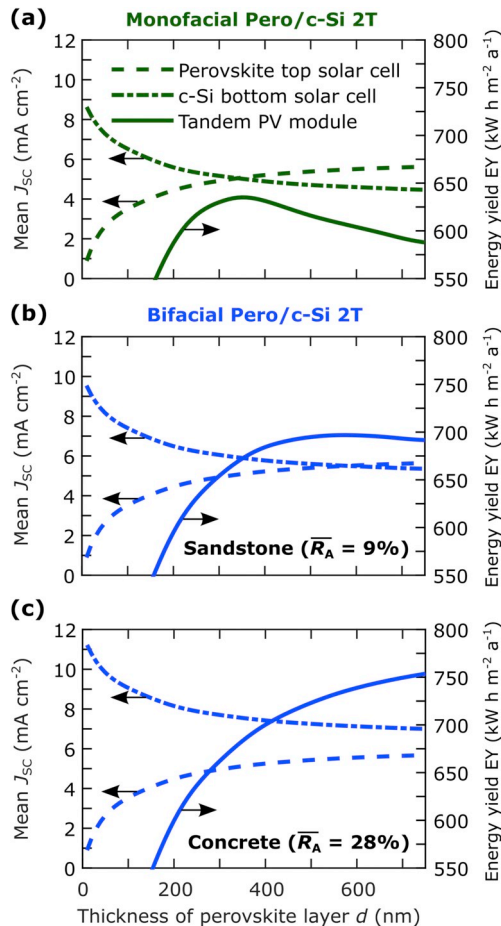


Fig. 4. Influence of albedo on current matching point in bifacial perovskite/c-Si 2T tandem PV modules with a perovskite bandgap of 1.72 eV. (a) Reference monofacial perovskite/c-Si 2T tandem PV module at the location Daggett (California) with optimized module tilt angle. (b) Bifacial perovskite/c-Si 2T tandem PV module with ground sandstone and (c) ground concrete. The mean J_{SC} over the time span of a year for perovskite top solar cell and c-Si bottom solar cell as well as the corresponding EY of the tandem PV module is shown as a function of the perovskite absorber layer thickness.

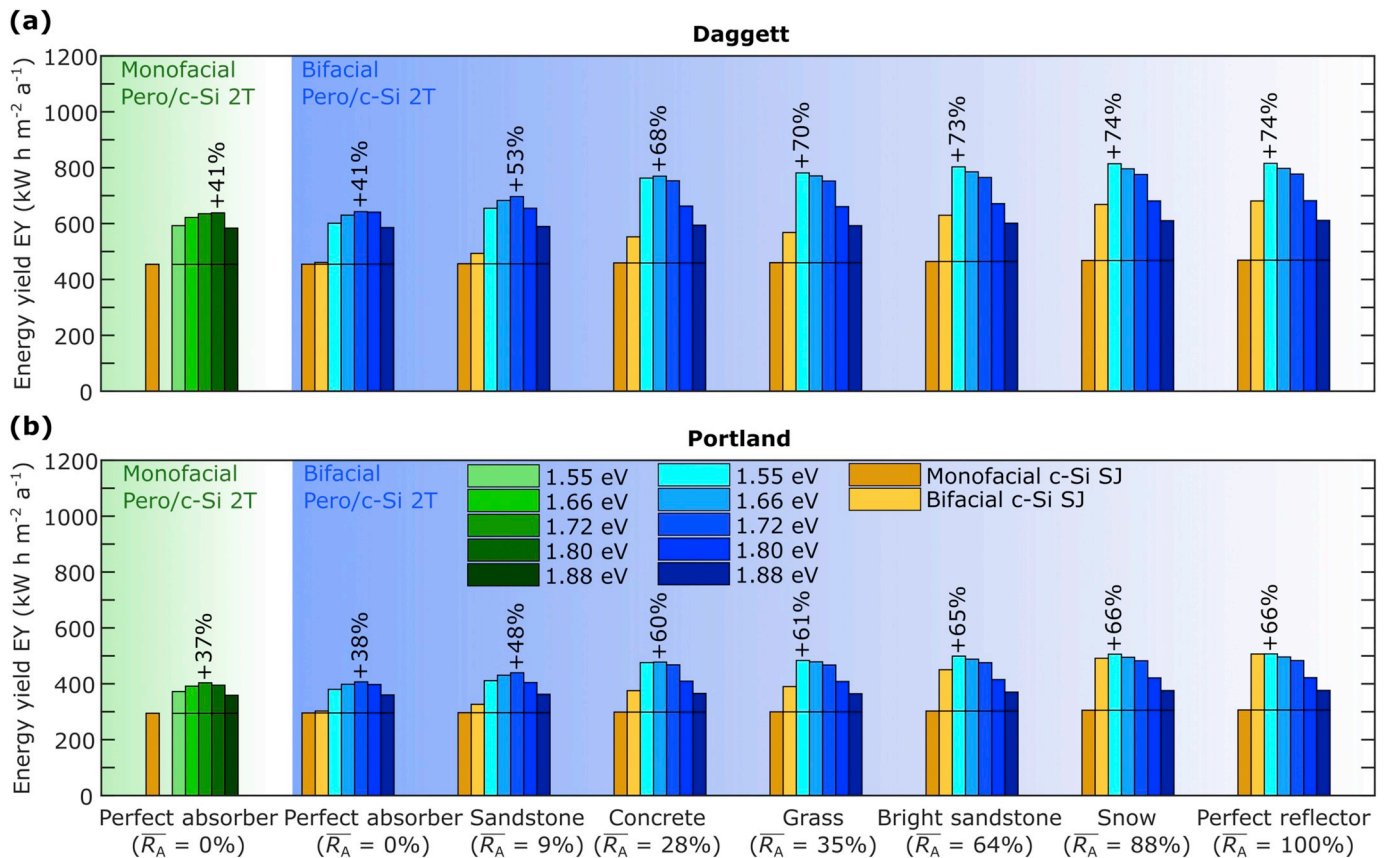


Fig. 5. Maximal EY of bifacial perovskite/c-Si 2T tandem PV modules for all investigated bandgaps of the perovskite as well as for five grounds, namely sandstone, concrete, grass, bright sandstone, and snow as well as two artificial grounds, perfect absorber and perfect reflector. The calculations are shown for the location (a) Daggett (California) and (b) Portland (Oregon) with optimized tilt angle. The arithmetic mean $\bar{R}_A$ of the corresponding albedo reflection spectra to ground is denoted. For all grounds, the relative enhancement in EY compared to the monofacial c-Si SJ PV module is indicated for the optimum bandgap with the highest EY. Additionally, we show the maximal EY of the three architectures monofacial perovskite/c-Si 2T tandem PV module, bifacial c-Si SJ PV module, and monofacial c-Si SJ PV module.

perovskite/c-Si 2T tandem PV module is calculated with respect to the perovskite absorber layer thickness in Fig. 4a for the location Daggett with hot desert climate. The calculations are carried out with a module tilt angle of 30°, which is close to the optimum tilt angle at location Daggett, in order to maximize EY. The investigated PV modules comprise metal-halide perovskites with a bandgap of 1.72 eV, since for monofacial perovskite/c-Si tandem PV modules, the optimal bandgap of perovskite is in the range of 1.7–1.8 eV [47]. The maximal EY is found

close to the perovskite thickness where the mean current densities match, as depicted in Fig. 4. The mean J_{SC} is the average in J_{SC} over the period of a year. With increasing albedo, we observe an enhanced charge carrier generation in the c-Si bottom solar cell (see Fig. 4b, c). This requires a thicker perovskite absorber layer in order to match the current density of both sub cells. In the present scenarios this implies that the optimal thickness of the perovskite absorber layer shifts from 350 nm to 570 nm and 750 nm for the grounds sandstone ($\bar{R}_A = 9\%$) and concrete

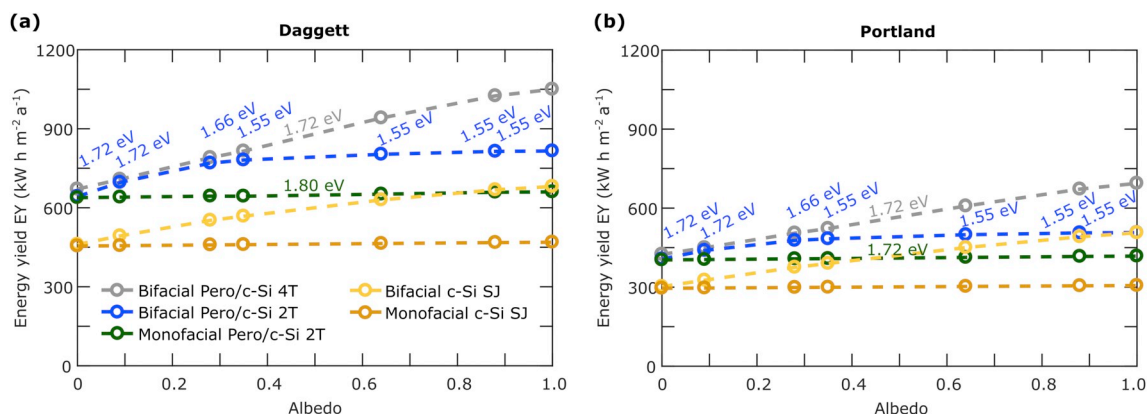


Fig. 6. Optimum EY for all investigated architectures of perovskite/c-Si tandem and c-Si SJ PV modules with respect to albedo at (a) Daggett and (b) Portland. The optimal bandgap of perovskite in all perovskite/c-Si tandem PV modules is indicated for all grounds.

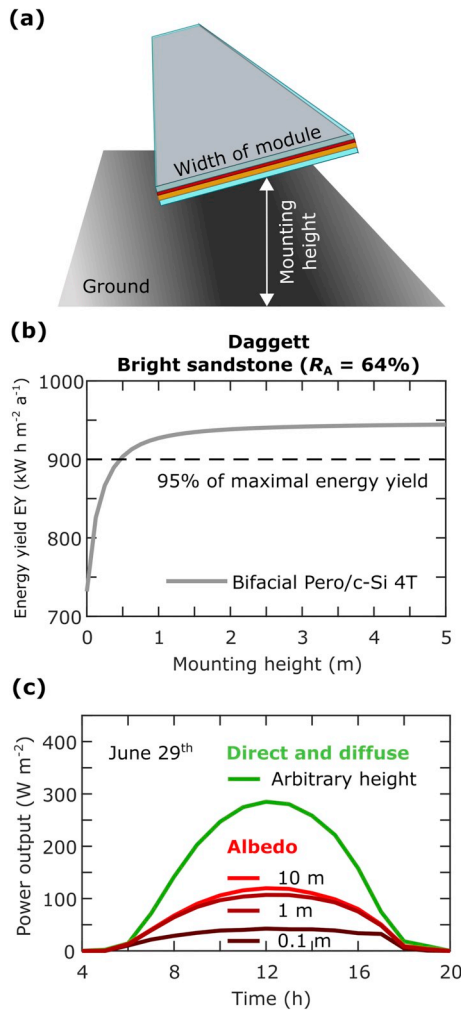


Fig. 7. (a) Scheme for a single row of PV modules elevated over ground and partly shading the ground. (b) EY as function of mounting height for bifacial perovskite/c-Si 4T tandem PV modules on the ground bright sandstone at location Daggett. (c) Corresponding power output during the course of a day for three mounting heights 0.1 m, 1 m, and 10 m. The power output is broken down into the parts related to direct and diffuse irradiance and related to albedo irradiance.

($\bar{R}_A = 28\%$) respectively. It is important to note that for the ground concrete with $\bar{R}_A$ of 28%, the optimal perovskite absorber layer thickness even exceeds the maximum thickness of 750 nm. In perovskite solar cells, the absorber layer thickness is commonly below 750 nm and hence we have chosen this as the upper limit of the perovskite absorber layer thickness in our simulation. The results shown in Fig. 4 further illustrate and quantify the relative enhancement in EY of 10% and 19% for bifacial perovskite/c-Si 2T tandem PV modules compared to monofacial perovskite/c-Si 2T tandem PV modules at the grounds sandstone ($\bar{R}_A = 9\%$) and concrete ($\bar{R}_A = 28\%$), respectively. While for low $\bar{R}_A$, the albedo light can be harvested entirely in a redesigned bifacial perovskite/c-Si tandem solar cell, for high $\bar{R}_A$ (i.e. ground concrete), no sufficient current matching is achieved by simply enhancing the perovskite absorber layer thickness. In the latter case, the bandgap of the perovskite top solar cell in the bifacial perovskite/c-Si tandem PV modules will need to be adapted to harvest the entire light incident at the rear side.

In order to overcome the before-mentioned constraints of current matching due to the limitation of current generation in the perovskite top solar cell (see Fig. 4), we determine the power output for different bandgaps of the perovskite. In Fig. 5, the effect of bandgap tuning on the

maximal EY is studied in detail. The maximal EY of bifacial perovskite/c-Si 2T tandem PV modules with optimized tilt angle is calculated as a function of the perovskite bandgap ranging from 1.55 eV to 1.88 eV, and at the same time as a function of the ground with $\bar{R}_A$ ranging from 0 to 100%. We note that the full albedo spectrum was taken into account. For this purpose, the range of albedo is complemented with two artificial grounds, a perfect absorber of constant 0% ground reflection as well as a perfect reflector of 100%. The mean albedo $\bar{R}_A$ for all grounds is specified in Fig. 5.

For a bifacial perovskite/c-Si 2T tandem PV module, we find a decrease of the optimal perovskite bandgap with increasing albedo. This trend persists for diverse climate conditions, and is exemplarily shown for the location Daggett (California) with a high direct part of irradiance (see Fig. 5a) as well as for Portland (Oregon) with a pronounced diffusive part (see Fig. 5b). It is worth mentioning that grounds as grass and snow do not occur at Daggett, but are shown here for the sake of completeness. Since the current generation in the c-Si bottom solar cell increases with albedo and therefore the current density is limited by the perovskite top solar cell (see Fig. 4), the use of low bandgap perovskites facilitates current matching for high albedos. In order to highlight the optimal perovskite bandgap for every ground, we indicate the maximal relative enhancement in EY compared to monofacial c-Si SJ PV modules in Fig. 5. Additionally, the simulated maximal EY of reference bifacial c-Si SJ PV modules is presented for all grounds. The prevalent installation site of bifacial PV is at grounds with moderate albedo intensity. As shown in Fig. 5, the optimal bandgap of metal-halide perovskites gradually shifts from 1.72 eV at ground sandstone ($\bar{R}_A = 9\%$) to 1.55 eV at ground grass ($\bar{R}_A = 35\%$). This highlights the importance of selecting the suitable bandgap to ground in the application of bifacial perovskite/c-Si 2T tandem PV modules.

Our results show clearly the influence of $\bar{R}_A$ on the overall device performance of bifacial perovskite/c-Si 2T tandem PV modules. It should be noted that this interrelation is also valid for alternative architectures of bifacial perovskite/c-Si 2T tandem PV modules with different light management schemes as referred to in the introduction (i – iv) (see Fig. S4, Supporting Information). We do not observe any impact of spectral ground reflection on the EY of bifacial PV modules with optimized tilt angle, since albedo irradiance mainly enters the PV module at the rear and hence enhances J_{SC} of c-Si bottom solar cell. Therefore, we expect a negligible influence of spectral changes in the incident sky irradiance, e.g., spectral shifts in the morning and evening. Furthermore, we find no effects of different ground reflection spectra on the EY, even for grass with a pronounced reflection at wavelengths above 700 nm (see Fig. 1).

In order to study both electrical configurations, which are currently of major interest, we compare the EY for 2T and 4T configurations. Both, the 2T exhibiting minimal number of electrodes and less parasitic absorption, and the 4T featuring independent operation of the sub-cells, surpass the EY of c-Si SJ PV modules. In Fig. 6, the maximal EY is shown for bifacial perovskite/c-Si 2T and 4T tandem PV modules as well as for the monofacial reference. The mean albedo intensity of all investigated grounds ranges from 0 – 100%. Bifacial perovskite/c-Si 2T and 4T tandem PV modules achieve a clearly enhanced EY with respect to bifacial c-Si SJ PV modules. The relative enhancement in EY of a bifacial perovskite/c-Si 2T tandem PV module is 24–38% for the ground grass ($\bar{R}_A = 35\%$). In bifacial perovskite/c-Si 2T tandem PV modules, the produced current density is limited due to current matching losses for albedo above 35%. This limitation is apparent by the saturation in EY (see Fig. 6) and originates from insufficient current generation in the perovskite top solar cell even for largest thicknesses. Since the 4T configuration does not suffer from constraints of current matching, bifacial perovskite/c-Si 4T tandem PV modules provide distinctly higher EY at grounds with high albedo, e.g., bright sandstone and snow. The perovskite bandgap for the optimum EY is depicted for all perovskite/c-Si tandem PV modules in Fig. 6, showing a substantial difference in the

optimal perovskite bandgap with regard to the architecture. This underlines the necessity for the optimal design of perovskite/c-Si tandem PV modules, to adapt the perovskite bandgap to the architecture. In case of 2T tandem PV modules, the optimal perovskite bandgap additionally depends on the albedo intensity.

The calculation of EY for different locations is also used to discuss the influence of the illumination spectrum on the optimal bandgap of metal-halide perovskites in perovskite/c-Si tandem PV. The changes in the overall spectrum of sky irradiance slightly affect the optimal bandgap, as shown by comparing the locations Daggett and Portland with very different climate conditions in Fig. 6. Accordingly, due to the different average photon energy (APE) of sky irradiance for Daggett with APE of 1.51 eV and Portland with 1.49 eV, the optimal bandgap of all investigated perovskite/c-Si 2T tandem PV modules is found at slightly lower bandgaps for Portland. The impact of the change in APE on the design of tandem PV modules was already intensively studied in earlier work on perovskite/CIGS tandem PV modules [31]. Our results highlight this aspect in the context of bifacial perovskite/c-Si tandem solar cells. The design of tandem perovskite/c-Si PV modules has to be adapted to the local climate conditions with regard to the bandgap in order to achieve optimal light harvesting.

The results presented so far depict the ideal case of a PV module with infinite mounting height. Our EY modelling allows for considering partial and directional shading of the ground, e.g., due to the shadow of the solar module itself. In earlier studies, the influence of ground shading on the performance of bifacial PV modules was investigated, addressing modules with different size, formation, and packing [24,42,48]. We determine the loss by ground shading for an infinite long single row of PV modules. The mounting height and width of the PV module are specified in Fig. 7a. In order to study the influence of ground shading on the device performance, we determine the EY of bifacial perovskite/c-Si 4T tandem PV modules for various mounting heights in the range of 0–5 m and with a fixed module width of 1 m. The overall EY as a function of the mounting height is exemplarily shown in Fig. 7b for the location Daggett at a ground with bright sandstone exhibiting $\overline{R_A}$ of 64%. For a mounting height of 0.4 m and higher, the calculated EY sums up to 95% of the ideal EY without any ground shadow. Since the overall losses in EY due to ground shading scale with the albedo intensity, these losses are even lower for common grounds ($\overline{R_A} < 64\%$). In Fig. 7c, we show the power output in EY throughout a day with respect to the mounting height. In order to illustrate the origin of performance losses, the power output related to albedo irradiance is separated from the power output related to direct and diffuse irradiance. With decreasing mounting height, the power output regarding to albedo irradiance decreases, whereas the power output regarding to direct and diffuse irradiance remains constant. It should be noted, that there are further relevant installation scenarios for bifacial PV modules beyond this study, demonstrating the complexity of module shading, e.g., shading due to frames, pillars, and surrounding objects. In particular, self-shading losses owing to adjacent PV modules are not considered in this study. Though, the problem of shading loss in bifacial PV is tackled in other literature as well as other numerical models [42,48–51].

4. Conclusion

In this work we have analyzed the EY of bifacial perovskite/c-Si tandem PV modules and have quantified the gain compared to opaque perovskite/c-Si tandem PV modules for various types of grounds. We found a significant gain in EY for bifacial perovskite/c-Si 2T tandem PV modules of 18–23% for the common ground grass (mean albedo = 35%) compared to monofacial perovskite/c-Si 2T tandem PV modules, and 24–38% compared to bifacial c-Si SJ PV modules. Furthermore, the prevalent electrical configurations, namely 2T and 4T, of bifacial perovskite/c-Si tandem PV modules have been investigated. For common grounds exhibiting albedo intensities in the range of 0–35%, both,

the 2T and 4T configuration outperform bifacial c-Si SJ PV modules in a similar extent with relative increase of 24–46%. For grounds with albedo above 35%, occurring rather rarely, the 4T configuration with independently connected sub-cells achieves particularly high EY. For the bifacial perovskite/c-Si 2T tandem PV module with a monolithic interconnection of the sub-cells and thus a minimal number of inverters, maximal power output requires careful device optimization, rather depending on the ground as on the location. Accordingly, our study demonstrates the considerable shift of the optimal bandgap of perovskite with albedo of the ground. Hence, selecting a suitable bandgap of the perovskite corresponding to the ground is of key importance for maximal EY of perovskite/c-Si 2T tandem PV modules.

Although bifacial PV modules possibly suffer from reduced albedo irradiance owing to ground shading, we found insignificant optical losses (<5%), if the module is elevated higher than 1 m above ground. The resulting gain in EY for all investigated bifacial PV modules compared to the monofacial PV modules is shown for all grounds with mean albedo ranging from 0 to 100%. Overall, this work demonstrates the potential of bifacial perovskite/c-Si tandem PV modules in order to enhance the power output by means of EY modelling considering realistic outdoor conditions.

Author contributions

Jonathan Lehr: Conceptualization, Methodology, Software, Validation, Investigation, Writing - Original Draft. **Malte Langenhorst:** Methodology, Software, Data Curation, Writing - Review & Editing. **Raphael Schmager:** Methodology, Software, Data Curation, Writing - Review & Editing. **Fabrizio Gota:** Methodology, Software, Writing - Review & Editing. **Simon Kirner:** Validation, Writing - Review & Editing. **Uli Lemmer:** Resources, Writing - Review & Editing, Supervision, Project administration, Funding acquisition. **Bryce S. Richards:** Resources, Writing - Review & Editing, Funding acquisition. **Chris Case:** Conceptualization, Validation, Writing - Review & Editing, Supervision. **Ulrich W. Paetzold:** Conceptualization, Software, Resources, Writing - Review & Editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

None.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.solmat.2019.110367>.

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